

Description of Work (DOW) for:

Full Stack Software Development Services (SMA2) [Cloud Service Support]

1.1 Objectives

CBO relies on modern, maintainable web-based applications and data tools to support its internal workflows, analytical pipelines, and reporting capabilities. To sustain and advance these capabilities, CBO seeks support with the ongoing development, enhancement, and maintenance of applications within its technology portfolio. The objective is to deliver reliable, secure, and scalable software solutions.

1.2 Scope of Work

CBO seeks professional full stack software development services to support the design, development, testing, and maintenance of web applications and supporting backend services. This engagement is focused on continuous delivery of working software through iterative development cycles, adherence to modern engineering practices, and close collaboration with CBO technical staff and stakeholders.

The Contractor shall provide development expertise across the full application stack, including Python-based backend APIs (FastAPI), ReactJS-based front-end interfaces, PostgreSQL databases, and containerized deployments targeting AWS cloud infrastructure. The Contractor shall operate within CBO's trunk-based development workflow, contributing clean, reviewed, and well-documented code through shared branching strategies with frequent integration to the main branch.

The Contractor shall perform all development and maintenance tasks necessary to sustain and advance CBO's application portfolio. This includes active contribution to ongoing sprint cycles, participation in planning and retrospective ceremonies, and delivery of production-ready software on a continuous basis.

The Contractor shall operate in close coordination with CBO's internal engineering team, following established coding standards, security baselines, and development workflows. All code produced under this contract shall be subject to peer review prior to merge, shall conform to agreed-upon style and linting standards, and shall not introduce known security vulnerabilities.

All development work shall be performed within CBO's trunk-based development model. The Contractor shall commit working, tested code frequently, maintain short-lived branches, and avoid long-running feature branches that create integration risk. Feature flags shall be used where appropriate to decouple deployment from release.

Upon completion of each sprint or development cycle, the Contractor shall participate in a structured review with CBO stakeholders to demonstrate completed work, document outcomes, and identify items for subsequent cycles.

The scope of work includes, but is not limited to, the following activities:

Application Development and Feature Delivery

- Design, develop, test, and deploy new features and enhancements across front-end (ReactJS) and back-end (Python/FastAPI) layers of CBO applications.
- Build and maintain RESTful APIs using FastAPI, ensuring proper validation, error handling, versioning, and documentation (e.g., OpenAPI/Swagger).
- Develop responsive, accessible front-end interfaces using ReactJS, adhering to CBO UX standards and Section 508 accessibility requirements.
- Implement and manage data models, queries, migrations, and stored procedures within PostgreSQL databases, with attention to performance and data integrity.

Trunk-Based Development and Collaboration

- Operate within CBO's trunk-based development workflow, committing small, incremental changes frequently to the main branch to reduce integration risk and maintain a releasable codebase.
- Participate in code reviews, providing and receiving constructive technical feedback to maintain code quality and consistency.
- Collaborate with CBO engineers, product owners, and analysts to understand requirements, refine user stories, and deliver working software iteratively.
- Adhere to CBO's branching, commit, and merge standards, including the use of short-lived feature branches and feature flags where applicable.

Containerization and Cloud Deployment

- Package applications and services using Docker, maintaining clean, minimal, and reproducible container images.
- Support deployment of containerized workloads to AWS environments (e.g., ECS, EKS, or equivalent managed services), working within existing cloud infrastructure patterns established by CBO's infrastructure team.
- Demonstrate working knowledge of AWS services relevant to application hosting and operation, including but not limited to EC2, S3, RDS, CloudWatch, IAM, and VPC networking concepts.
- Collaborate with CBO infrastructure staff on environment configuration, secrets management, and cloud resource usage. The Contractor is not expected to serve as a dedicated IaC or CI/CD engineer but should be comfortable working within automated deployment pipelines and contributing pipeline-compatible code.

Maintenance, Bug Fixes, and Technical Debt Reduction

- Investigate, diagnose, and resolve application defects, performance issues, and security vulnerabilities across all layers of the stack.
- Proactively identify and address technical debt, refactoring code to improve maintainability, readability, and test coverage.

- Monitor application behavior and logs (e.g., via AWS CloudWatch or equivalent) and respond to operational issues as needed.

Testing and Quality Assurance

- Write and maintain automated unit, integration, and regression tests for both back-end (pytest or equivalent) and front-end (Jest, React Testing Library, or equivalent) code.
- Ensure new features and bug fixes are accompanied by appropriate test coverage prior to merge.
- Participate in functional and user acceptance testing in coordination with CBO stakeholders.

Documentation

- Maintain accurate, up-to-date technical documentation for all developed systems, including API documentation, data model diagrams, architecture decision records (ADRs), and developer setup guides.
- Document operational procedures, environment configurations, and known issues in CBO's designated knowledge management platform.

1.3 Key Personnel

The following role is anticipated to be required for this work and is considered Key Personnel.

- Full Stack Software Developer

1.3.1 Key Personnel Requirements

The Contractor shall provide a Full Stack Developer who meets the following specific requirements:

- Demonstrated proficiency in the core technology stack: Python, FastAPI, ReactJS, PostgreSQL, and Docker.
- Possess sufficient familiarity with AWS cloud services to deploy, monitor, and troubleshoot applications in cloud-hosted environments.
- Knowledge of CI/CD pipeline concepts and IaC tooling (e.g., Terraform, GitHub Actions) is desired but is not a primary responsibility of this role; the Contractor is expected to write code that integrates cleanly into existing automated pipelines.

1.4 Deliverables

Phase 1: Onboarding and Environment Setup – Target duration of 2 weeks

- Complete access provisioning, environment setup, and orientation to CBO's codebase, development standards, and tooling.
- Review existing application architecture, database schemas, and deployment configurations.
- Deliver an onboarding summary documenting system understanding, identified gaps or risks, and a proposed initial sprint backlog contribution.

Task	Type	Due Date / Frequency
Environment setup and access confirmation	Milestone	End of Week 2
Onboarding summary and initial backlog review	Deliverable	End of Week 2

Phase 2: Active Development and Maintenance – Ongoing

Ongoing sprint-based delivery of features, enhancements, bug fixes, and documentation in accordance with CBO's development roadmap and priorities.

Task	Type	Due Date / Frequency
Sprint deliverables (features, fixes, tests)	Deliverable	Per sprint cadence (bi-weekly)
Code review participation	Recurring	Ongoing
Development status report	Recurring Report	Weekly
Updated technical documentation	Deliverable	With each significant feature or change

Phase 3: Testing and Quality Validation – Ongoing

Continuous test development and quality assurance activities embedded within each sprint cycle.

Task	Type	Due Date / Frequency
Unit and integration test delivery	Deliverable	With each feature/fix
Test coverage summary	Recurring Report	Bi-weekly, at sprint close
Bug resolution report	Recurring Report	Weekly

Phase 4: Documentation and Knowledge Transfer – Target duration of 1–2 weeks at engagement close

- Deliver complete and current technical documentation for all systems worked on under this contract.
- Conduct structured knowledge transfer sessions with CBO engineering staff.
- Submit a final summary report covering all completed work, outstanding items, and recommendations.

Task	Type	Due Date / Frequency
Final technical documentation package	Deliverable	End of engagement
Knowledge transfer sessions	Deliverable	End of engagement
Final summary and transition report	Deliverable	End of engagement

1.5 Acceptance Criteria

In association with SOW paragraph 2.6.2, the project shall be deemed successful upon:

- Continuous delivery of working, tested software meeting agreed sprint acceptance criteria
- Completion of all documentation and knowledge transfer activities
- Approval by CBO IRM and COR
- Positive assessment of code quality, test coverage, and adherence to CBO development standards

1.6 Performance Metrics & Compliance

In addition to the requirements of SOW paragraph 2.6.3, the Contractor shall ensure that all software development activities align with applicable federal standards, including NIST SP 800-53 Rev. 5, NIST SP 800-171, and FISMA. Performance will be evaluated based on:

- Velocity and quality of sprint deliverables
- Defect rates and resolution times
- Test coverage levels maintained across the codebase
- Adherence to trunk-based development practices and code review participation
- Compliance with CBO security and coding standards
- System uptime and application stability

Capability	NIST SP 800-53 Control Families	NIST SP 800-171 Requirements
Secure API Development (FastAPI)	SC, AC	3.1 Access Control, 3.13 System & Communications Protection
Authentication & Authorization	IA, AC-2, AC-6	3.5 Identification & Authentication
Data Protection (PostgreSQL)	MP, SC-28	3.8 Media Protection, 3.13.16
Container Security (Docker/AWS)	CM, SI	3.4 Configuration Management, 3.14 System Integrity
Logging & Monitoring (CloudWatch)	AU, IR	3.3 Audit & Accountability, 3.6 Incident Response
Secure Coding Practices	SA, SI-10	3.14 System Integrity
Access to Cloud Resources (AWS IAM)	AC, IA, SC	3.1, 3.5, 3.13

1.7 *Government Furnished Equipment and Information*

The Government will provide the following to perform the tasks outlined in this Description of Work:

- CBO-issued laptops, mobile devices, and peripheral hardware (if required)
- Access to internal systems for testing access to tools such as Momentum, Microsoft 365, SharePoint, Zoom, Cisco Webex, and VPN credentials
- Network diagrams, system architecture documentation, current firewall rule sets, and relevant configuration files
- Applicable policies, procedures, and security baselines (e.g., NIST, FISMA, IRS 1075 guidelines)
- Contact information for internal technical points of contact (POCs) and data center operations